

Optical RIS-Aided Covert Visible Light Communications

Lei Qian¹, Fangqian Wu, Di Wang², Nuo Huang¹,
and Zhengyuan Xu¹, *Senior Member, IEEE*

Abstract—This paper investigates a covert visible light communication (VLC) system, where a transmitter (Alice) aims to send confidential information to a legitimate user (Bob) while avoiding being detected by an unauthorized warden (Willie). To achieve this, we propose an optimization framework that controls both reflection by mirror array-based reconfigurable intelligent surface (mRIS) and refraction by liquid crystal-based RIS (LC-RIS) to reconstruct a desired covert optical transmission environment. Considering the covertness constraint with the strongest Willie aided by LC-RIS, we jointly optimize the rotation configuration of mRIS, the parameter of Bob's LC-RIS, and the average optical intensity of Alice by a proposed stepwise decoupling method. The simulation results show that the deployment of LC-RIS on Bob's receiver has a more significant impact on enhancing covert VLC performance compared to the deployment of mRIS.

Index Terms—Covert communications, visible light communication, reconfigurable intelligent surface, mirror array.

I. INTRODUCTION

Considering the natural advantage of transmission security, visible light communication (VLC) is expected to be deployed in confidential communication scenarios. However, due to the broadcast nature of visible light, VLC still has the risk of hijacking and bypass listening. Some existing works aim to improve the security of VLC by physical layer security (PLS) techniques, such as digital beamforming and artificial noise [1]. By exploiting and enlarging the difference between legitimate channels and wiretap channels, PLS techniques can prevent eavesdroppers from decoding confidential information in the received signals [2]. However, when the malicious eavesdroppers collude to improve their decoding ability or utilize advanced channel state information estimation, the security guarantee provided by PLS will be degraded [3].

As a higher level security technique, covert communications can prevent privacy leakage essentially by hiding transmitted signals [4]. Without even being aware of the existence of the transmission, wardens will not try to decode the content in the signals [5]. For an unmanned aerial vehicle-based covert VLC system, a height optimization is proposed based on the analysis of the covertness constraint of the detection channel in [6].

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Lei Qian, Fangqian Wu, and Di Wang are with the Tianjin Key Laboratory of Optoelectronic Detection Technology and System, School of Electronic and Information Engineering, Tiangong University, Tianjin 300160, China (e-mail: qianlei@tiangong.edu.cn; 2331081035@tiangong.edu.cn; wangdi@tiangong.edu.cn).

Nuo Huang and Zhengyuan Xu are with the CAS Key Laboratory of Wireless-Optical Communications, School of Information Science and Technology, University of Science and Technology of China, Hefei 230027, China (e-mail: huangnuo@ustc.edu.cn; xuzhy@ustc.edu.cn).

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As a promising solution for 6G, reconfigurable intelligent surface (RIS) [7], [8] opens up a new horizon for covert transmission enhancement by restructuring wireless propagation environments. In VLC systems, there are three types of optical RIS. The first one is the mirror array implemented RIS (mRIS), which can manipulate the reflection environment by adjusting the rotation of mirrors [1]. The second one is the liquid crystal-based RIS (LC-RIS), which can control the refraction of light signals by changing the arrangement of liquid crystal molecules electrically [9]. The third one is the programmable meta-surface reflector realized RIS, which can change the phase gradients of incident waves by using the synthesized material [10]. Considering the expensive cost of the programmable meta-surface, mRIS and LC-RIS are the two mainly used optical RIS in the current study. The static frequency flat channel of the mRIS-aided VLC system is investigated in [11]. Further, considering mirror elements at different spatial positions may observe different channels, the spatial non-stationary channel is studied in [12]. Based on the irradiance analysis in [11], the potential of VLC security enhancement provided by mRIS is first explored in [13]. Then, the mRIS-aided PLS optimization is studied in [14] for the multi-user VLC systems. Besides, the required voltage of LC-RIS-aided VLC are investigated in [15]. Further, jointly optimizing mRIS and LC-RIS, achievable rate maximization methods are proposed in [9] and [16].

Motivated by the above discussions, this paper considers an optical RIS-aided covert VLC system including a transmitter, a legitimate user and a warden. With an mRIS deployed on the wall and LC-RIS equipped on receivers, we propose an optimization framework for covert VLC to manipulate the optical transmission environment in both reflection and refraction. First, we optimize the refraction index for warden's LC-RIS to model the covertness constraint in the worst case of covert VLC. Then, from the system-level perspective, we propose a stepwise decoupling method to jointly optimize the rotation configuration of mRIS, the refraction index of legitimate user's LC-RIS and the power of the transmitter. Finally, numerical results verify the theoretical analysis.

Notations: In this paper, matrices and scalars are represented using bold and normal fonts, respectively. In particular, the symbol $\mathcal{V}_T(\cdot, \cdot)$ represents the total variance distance, while $\mathcal{D}(\cdot||\cdot)$ denotes the relative entropy; the symbol $\sum$ is employed to indicate the summation; the symbol $\mathcal{N}$ represents the normal distribution; the symbol $\mathbb{E}(\cdot)$ denotes the expectation operator.

II. SYSTEM MODEL

We consider an optical RIS-aided covert VLC system consisting of a transmitter (Alice), a legitimate user (Bob), and a warden (Willie), as shown in Fig. 1. Alice is deployed on the ceiling of the room, while Bob is placed on the receiving plane. The goal of Willie is to detect any potential covert communication between Alice and Bob. To characterize different cases of Willie's detection, we consider that Willie is moving on the receiving plane. Adopting a light-emitting diode (LED) array, Alice transmits optical signals by utilizing a direct current-biased intensity modulation/direct detection (IM/DD) scheme. In addition, Bob and Willie each employ a photodiode (PD) to receive and detect optical signals.

In the covert VLC system, Alice transmits the stealthy message to Bob, avoiding the detection by Willie. Both Bob and Willie equip LC-RIS-aided receivers, which are realized by an LC-based RIS module placed in front of the PD. The refractive index of the LC-based RIS

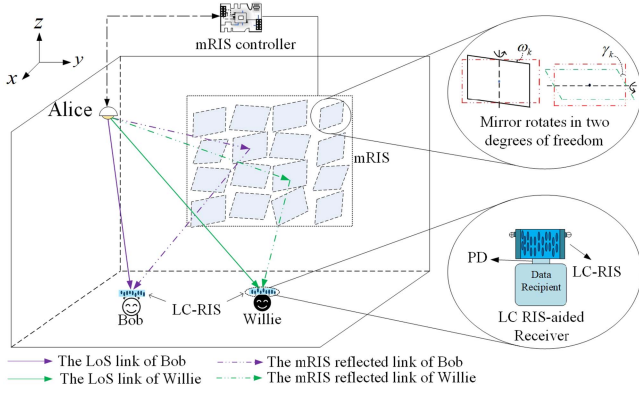


Fig. 1. An optical RIS-aided covert VLC system.

module can be controlled by an external electric field to achieve light steering and amplification capabilities.

Besides, mRIS is introduced to build a controllable reflection environment. As shown in Fig. 1, a two-dimensional mRIS consisting of N_m identical rectangular mirrors is installed on one wall. Driven by two servos, each mirror can be rotated independently with two degrees of freedom. To jointly control the operations of servos of different mirrors, an mRIS controller with N_s servos control boards is used and the bus control protocol is adopted [17]. By orchestrating N_s control boards in parallel, each mirror can reach the desired rotation within a defined period. The rotation of mirror k is determined by two clockwise angles, namely the roll angle γ_k and yaw angle ω_k . We define $\boldsymbol{\gamma} \in \mathbb{R}^{N_m \times 1}$ and $\boldsymbol{\omega} \in \mathbb{R}^{N_m \times 1}$ as the collection of $\{\gamma_k\}$ and $\{\omega_k\}$, where $k = 1, 2, \dots, N_m$.

A. Received Signals Over Communication Channel

Alice sends an optical signal vector $\mathbf{x} = \{x[i]\}_{i=1}^N$ to Bob over N channels, where $x[i]$ denotes the signal over the i -th channel. Considering non-negativity, peak and average optical power constraints, $x[i]$ should satisfy

$$\begin{cases} 0 \leq x[i] \leq A, & \forall i \in \{1, 2, \dots, N\}, \\ \mathbb{E}[x[i]] = \xi P, & \forall i \in \{1, 2, \dots, N\}, \end{cases} \quad (1)$$

where A and P denote the peak and average optical intensity of Alice, respectively; and $\xi \in [0, 1]$ signifies the dimming target. The additive white Gaussian noise (AWGN) vector at Bob in N channels is denoted by $\mathbf{n}_B = \{n_B[i]\}_{i=1}^N$, where $n_B[i] \sim \mathcal{N}(0, \sigma_B^2)$ and σ_B^2 is the noise variance at Bob. Under the communication channel, the vector of signals received by Bob is denoted by $\mathbf{y}_B = \{y_B[i]\}_{i=1}^N$, where $y_B[i]$ denotes the signal received over the i -th channel, which is expressed as

$$y_B[i] = h_B^{LC} \left(h_B^{LoS} + \sum_{k=1}^{N_m} h_{B,k}^{mRIS} \right) x[i] + n_B[i], \quad (2)$$

where h_B^{LoS} denotes the line-of-sight (LoS) channel gain between Alice and Bob, h_B^{LC} represents the channel gain provided by LC-RIS-aided receiver of Bob, $h_{B,k}^{mRIS}$ denotes the mRIS channel gain of Bob contributed by mirror k . The channel gains will be modeled in Section III.

B. Received Signals Over Detection Channel

On the detection channel, Willie uses statistical hypothesis test to determine whether Alice is communicating. The vector of signals detected by Willie in N channels is denoted by $\mathbf{y}_W = \{y_W[i]\}_{i=1}^N$,

where $y_W[i]$ represents the signal over the i -th channel. The AWGN vector at Willie in N channels is denoted by $\mathbf{n}_W = \{n_W[i]\}_{i=1}^N$, where $n_W[i] \sim \mathcal{N}(0, \sigma_W^2)$ and σ_W^2 is the noise variance at Willie.

According to the binary hypothesis test performed by Willie, the null hypothesis $\mathcal{H}_0$ means that Alice is not communicating to Bob; the alternative hypothesis $\mathcal{H}_1$ means that Alice is sending a message to Bob. Hence, $y_W[i]$ can be expressed as

$$\begin{cases} \mathcal{H}_0 : & y_W[i] = n_W[i], \\ \mathcal{H}_1 : & y_W[i] = h_W^{LC} \left(h_W^{LoS} + \sum_{k=1}^{N_m} h_{W,k}^{mRIS} \right) x[i] + n_W[i], \end{cases} \quad (3)$$

where h_W^{LoS} denotes the LoS detection channel gain, h_W^{LC} represents the channel gain provided by LC-RIS-aided receiver of Willie, and $h_{W,k}^{mRIS}$ denotes the mRIS channel gain of Willie contributed by mirror k .

Based on the average power of the detected signal, Willie's decision rule is described as

$$\begin{aligned} & \mathcal{D}_1 \\ & \bar{p}_W \gtrless \Omega, \\ & \mathcal{D}_0 \end{aligned} \quad (4)$$

where $\bar{p}_W = \frac{1}{N} \sum_{i=1}^N |y_W[i]|^2$ represents the average power of the detected signal; $\Omega > 0$ is the detection threshold; $\mathcal{D}_0$ denote the decision in favor of $\mathcal{H}_0$, which is made if $\bar{p}_W < \Omega$; $\mathcal{D}_1$ denote the decision in favor of $\mathcal{H}_1$, which is made if $\bar{p}_W \geq \Omega$.

The detection performance at Willie is typically evaluated by the detection error probability, which is given by

$$P_e = P[\mathcal{H}_0] P[\mathcal{D}_1|\mathcal{H}_0] + P[\mathcal{H}_1] P[\mathcal{D}_0|\mathcal{H}_1], \quad (5)$$

where $P[\mathcal{D}_1|\mathcal{H}_0]$ and $P[\mathcal{D}_0|\mathcal{H}_1]$ denote the probability of false alarm error and missed detection, respectively. Since Willie's decisions about Alice's transmission state are akin to a random guess [18], [19], [20], we assume that each hypothesis has an equal priori probability, i.e., $P[\mathcal{H}_0] = P[\mathcal{H}_1] = 0.5$.

To achieve covert transmission, Alice prefers a high value of P_e , which means that even if Willie employs an optimal detector, Alice's transmission remains undetectable. Therefore, the covertness constraint is modeled as $P[\mathcal{D}_1|\mathcal{H}_0] + P[\mathcal{D}_0|\mathcal{H}_1] \geq 1 - \varepsilon$, where ε is a small constant.

III. CHANNEL DESCRIPTION

A. LoS Channel Gain

The LoS channel gain for receiver j ($j = B$ for Bob, $j = W$ for Willie) is expressed as [9]

$$h_j^{LoS} = \begin{cases} \frac{(m+1)\varpi\mu T\vartheta a^2}{2\pi d_j^2 \sin^2(\psi_j^s)} \cos^m(\varphi_j^s) \cos(\psi_j^s), & \text{if } 0 \leq \psi_j^s \leq \psi_F, \\ 0, & \text{if } \psi_j^s \geq \psi_F, \end{cases} \quad (6)$$

where d_j is the distance from Alice to receiver j , ϑ is the physical area of the PD, m is the Lambertian coefficient, μ , ϖ , T and a are the electro-optical conversion efficiency, the responsivity of the PD, the gain of the transimpedance amplifier, and the refraction index, respectively, ψ_j^s and φ_j^s are the angles of incidence and irradiance of receiver j 's LoS link, respectively, and ψ_F is half of field-of-view (FoV).

B. mRIS Channel Gain

Assuming that the reflected links provided by mRIS can reach the receiver instantaneously, we model the mRIS channel as a static

frequency flat channel. For mirror k rotating with γ_k and ω_k , the mRIS channel gain of receiver j is expressed as [9]

$$h_{j,k}^{\text{mRIS}}(\gamma_k, \omega_k) = \begin{cases} \frac{\rho(m+1)\varpi\mu T\vartheta a^2}{2\pi(d_k^s)^2(d_j^k)^2\sin^2(\psi_j^k)} dA_k \cos^m(\theta_k^s) \cos(\psi_k^s) \\ \times \cos(\theta_j^k) \cos(\psi_j^k), & 0 \leq \psi_j^k \leq \psi_F, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where ρ is the specular reflection coefficient, d_k^s is the distance between Alice and mirror k , d_j^k is the distance between mirror k and receiver j , θ_k^s is the angle of irradiance from Alice to the reflective surface k , ψ_k^s is the angle of incidence on mirror k , θ_j^k is the angle of irradiance from mirror k towards receiver j , and ψ_j^k is the angle of incidence of the reflected signal from mirror k . The cosine of θ_j^k is expressed as [9]

$$\cos(\theta_j^k) = \frac{(x_k - x_j)}{d_j^k} \sin(\gamma_k) \cos(\omega_k) + \frac{(y_k - y_j)}{d_j^k} \cos(\gamma_k) \cos(\omega_k) + \frac{(z_k - z_j)}{d_j^k} \sin(\omega_k), \quad (8)$$

where (x_k, y_k, z_k) and (x_j, y_j, z_j) are the coordinates of mirror k and receiver j , respectively.

C. Channel Gain Provided by LC-RIS-Aided Receiver

For receiver j deployed with an LC-RIS, the channel gain provided by LC-RIS is expressed as

$$h_j^{\text{LC}} = \zeta_j \delta_j, \quad (9)$$

where ζ_j and δ_j represent the transition coefficient and the amplification coefficient of the LC-RIS.

Based on the relationship between the angular transmittance and the angular reflectance [15], ζ_j is expressed as

$$\zeta_j = (1 - R_{\text{in}}(\psi_j))(1 - R_{\text{out}}(\varphi_j)), \quad (10)$$

where ψ_j and φ_j denote angles of incidence and refraction on the LC-RIS, respectively, $R_{\text{in}}(\psi_j)$ and $R_{\text{out}}(\varphi_j)$ represent the amount of light that gets reflected when the light signal enters or exits the LC-RIS, respectively. Let η_a denote the refractive index in air and $\eta = \eta_j/\eta_a$ denote the ratio of the refractive index in air and LC-RIS. Then, we have [15]

$$R_{\text{in}}(\psi_j) = \frac{1}{2} \left[\frac{\eta^2 \cos(\psi_j) - \sqrt{\eta^2 - \sin^2(\psi_j)}}{\eta^2 \cos(\psi_j) + \sqrt{\eta^2 - \sin^2(\psi_j)}} \right]^2 + \frac{1}{2} \left[\frac{\cos(\psi_j) - \sqrt{\eta^2 - \sin^2(\psi_j)}}{\cos(\psi_j) + \sqrt{\eta^2 - \sin^2(\psi_j)}} \right]^2, \quad (11)$$

and

$$R_{\text{out}}(\varphi_j) = \frac{1}{2} \left[\frac{\eta^{-2} \cos(\varphi_j) - \sqrt{\eta^{-2} - \sin^2(\varphi_j)}}{\eta^{-2} \cos(\varphi_j) + \sqrt{\eta^{-2} - \sin^2(\varphi_j)}} \right]^2 + \frac{1}{2} \left[\frac{\cos(\varphi_j) - \sqrt{\eta^{-2} - \sin^2(\varphi_j)}}{\cos(\varphi_j) + \sqrt{\eta^{-2} - \sin^2(\varphi_j)}} \right]^2. \quad (12)$$

Besides, the amplification coefficient in (9) is expressed as

$$\delta_j = \exp\left(\frac{2\pi v_e \eta_j^3 \Gamma}{\lambda \cos(\psi_j)}\right), \quad (13)$$

where λ is the wavelength of the incident light, Γ is the electro-optic coefficient, and v_e is an externally applied voltage. The relationship between v_e and η_j is characterized by [16]

$$v_e = v_{\text{th}} - v_0 \ln \left[-\tan \left(\frac{\tan^{-1} \left(\frac{\eta_1 \sqrt{(\eta_2^2 - \eta_1^2)(\eta_2^2 - \eta_j^2)}}{\eta_j(\eta_2^2 - \eta_1^2)} \right)}{2} - \frac{\pi}{4} \right) \right]. \quad (14)$$

where v_{th} is the critical voltage threshold, v_0 is a constant determined by the materials of the LC-RIS, η_1 and η_2 are the upper and lower bounds on the refractive index.

IV. OPTIMIZATION OF THE OPTICAL RIS-AIDED COVERT VLC SYSTEM

A. Optimization on Willie's Side

Considering Willie hopes to obtain the best detection with the help of LC-RIS, we formulate a detection error probability minimization problem on Willie's side, which is expressed as

$$\text{OP1: } \min_{\eta_W} P_e \quad (15a)$$

$$\text{s.t. } \eta_1 \leq \eta_W \leq \eta_2 \quad (15b)$$

According to [4], P_e in (5) can be characterized by

$$P_e = \frac{1}{2} (P[\mathcal{D}_1|\mathcal{H}_0] + P[\mathcal{D}_0|\mathcal{H}_1]) = \frac{1}{2} (1 - \mathcal{V}_T(p_1(\mathbf{y}_W), p_0(\mathbf{y}_W))), \quad (16)$$

where $p_0(\mathbf{y}_W)$ denotes the probability density function (PDF) of the Willie's observations when Alice does not transmit information (i.e., $\mathcal{H}_0$ is true), $p_1(\mathbf{y}_W)$ denotes the PDF of the Willie's observations when Alice transmits information (i.e., $\mathcal{H}_1$ is true). Based on the Pinsker's inequality [4], we have

$$\mathcal{V}_T(p_1(\mathbf{y}_W), p_0(\mathbf{y}_W)) \leq \sqrt{\frac{\ln 2}{2} \mathcal{D}(p_1(\mathbf{y}_W) || p_0(\mathbf{y}_W))}, \quad (17)$$

According to (16) and (17), minimizing P_e is equivalent to maximizing $\mathcal{D}(p_1(\mathbf{y}_W) || p_0(\mathbf{y}_W))$. Based on the chain rule of relative entropy and the entropy-power inequality, the upper bound of $\mathcal{D}(p_1(\mathbf{y}_W[i]) || p_0(\mathbf{y}_W[i]))$ can be derived as (for brevity, we abbreviate $y_W[i]$ as y_W)

$$\mathcal{D}(p_1(y_W) || p_0(y_W)) \leq \frac{(4\pi - e) \left[h_W^{\text{LC}} \left(h_W^{\text{LoS}} + \sum_{k=1}^{N_m} h_{W,k}^{\text{mRIS}} \right) \xi P \right]^2}{4\pi \sigma_W^2 \ln 2} + \frac{\left[h_W^{\text{LC}} \left(h_W^{\text{LoS}} + \sum_{k=1}^{N_m} h_{W,k}^{\text{mRIS}} \right) \xi P \right]^4 e^2}{16\pi^2 \sigma_W^4 \ln 2}, \quad (18)$$

from which we can infer that maximizing $\mathcal{D}(p_1(\mathbf{y}_W) || p_0(\mathbf{y}_W))$ is equivalent to maximizing the detection channel gain.

Hence, we can transform OP1 into

$$\text{OP1-1: } \max_{\eta_W} h_W^{LC} \left(h_W^{LoS} + \sum_{k=1}^{N_m} h_{W,k}^{mRIS} \right) \quad (19a)$$

$$\text{s.t. } \eta_l \leq \eta_W \leq \eta_2 \quad (19b)$$

In light of the fact that the mRIS controller has been commissioned by Alice for the purpose of facilitating covert transmission, it is reasonable to assume that the mRIS under control will not provide channel gain for Willie, which means $h_{W,k}^{mRIS} = 0$ for all mirrors. By verifying the second order derivation of $h_W^{LC} h_W^{LoS}$ on $\eta_W \in [\eta_l, \eta_2]$, we obtain that OP1-1 is a convex problem, which can be solved quickly by the gradient descent method.

B. Optimization on System Level

For the system-level optimization, we try to maximize the achievable covert communication rate under the strictest covertness constraint. The lower bound in [16] is adopted as the achievable covert communication rate, i.e.,

$$R_B(\eta_B, \gamma, \omega, P) = \frac{1}{2} \log \left[1 + \frac{e \xi^2 P^2 (h_B^{LC})^2}{2\pi \sigma_B^2} \left(h_B^{LoS} + \sum_{k=1}^{N_m} h_{B,k}^{mRIS} \right)^2 \right]. \quad (20)$$

According to (16), we rewrite the covertness constraint as

$$\mathcal{D}(p_1(y_W) || p_0(y_W)) \leq \frac{2\varepsilon^2}{N \ln 2}. \quad (21)$$

Based on the optimization result on Willie's side, we define the covertness constraint with optimal η_W^* as $\mathcal{D}^*(p_1(y_W) || p_0(y_W)) \leq \frac{2\varepsilon^2}{N \ln 2}$, which characterizes the worst case of the covert VLC system.

The system-level optimization problem is formulated as

$$\text{OP2: } \max_{\eta_B, \gamma, \omega, P} R_B(\eta_B, \gamma, \omega, P) \quad (22a)$$

$$\text{s.t. } \mathcal{D}^*(p_1(y_W) || p_0(y_W)) \leq \frac{2\varepsilon^2}{N \ln 2} \quad (22b)$$

$$\eta_l \leq \eta_B \leq \eta_2 \quad (22c)$$

$$-\frac{\pi}{2} \leq \gamma_k \leq \frac{\pi}{2}, k = 1, 2, \dots, N_m \quad (22d)$$

$$-\frac{\pi}{2} \leq \omega_k \leq \frac{\pi}{2}, k = 1, 2, \dots, N_m \quad (22e)$$

where constraint (22c) characterizes the range of the refractive index of Bob's LC-RIS, constraints (22d) and (22e) characterize the rotation ranges of each mirror of mRIS.

In OP2, jointly optimizing the rotation configuration of mRIS, the parameter of Bob's LC-RIS and the average optical intensity of Alice makes the system-level optimization problem hard to solve directly. In this paper, by exploring the features of different variables, we propose a stepwise decoupling method to handle OP2. First, we fix variables η_B, P and transform OP2 into

$$\text{OP2-1: } \max_{\gamma, \omega} R_B(\gamma, \omega | P, \eta_B) \quad (23)$$

$$\text{s.t. } (22b), (22c), (22d), (22e).$$

Considering the non-convexity of R_B on γ and ω in OP2-1, we exploit the principle that the position of the reflected spot in the receiving

plane has a one-to-one mapping with the rotation angles, which is characterized by [13]

$$\begin{cases} \omega_k = \sin^{-1} \left(\hat{\mathbf{N}}_k^T(\mathbf{Q}) \mathbf{e}_3 \right), \\ \gamma_k = \sin^{-1} \left(\frac{\hat{\mathbf{N}}_k^T(\mathbf{Q}) \mathbf{e}_1}{\cos(\omega_k)} \right), \end{cases} \quad (24)$$

where $\mathbf{e}_1 = [1, 0, 0]^T$, $\mathbf{e}_3 = [0, 0, 1]^T$, $\mathbf{Q} = [x_q, y_q, z_q]^T$ denotes the position of the focused reflected spot and $\hat{\mathbf{N}}_k(\mathbf{Q})$ denotes the normal vector of mirror k after rotation towards $\mathbf{Q}$. According to Snell's law of reflection, $\hat{\mathbf{N}}_k(\mathbf{Q})$ can be expressed as

$$\hat{\mathbf{N}}_k(\mathbf{Q}) = \frac{(\mathbf{S} - \mathbf{R}_k) / \|\mathbf{S} - \mathbf{R}_k\|_2 + (\mathbf{Q} - \mathbf{R}_k) / \|\mathbf{Q} - \mathbf{R}_k\|_2}{\|(\mathbf{S} - \mathbf{R}_k) / \|\mathbf{S} - \mathbf{R}_k\|_2 + (\mathbf{Q} - \mathbf{R}_k) / \|\mathbf{Q} - \mathbf{R}_k\|_2\|_2}, \quad (25)$$

where $\mathbf{S}$ and $\mathbf{R}_k$ denote the position of Alice and the center position of mirror k , respectively. Based on the one-to-one mapping between γ and ω , we transform OP2-1 into

$$\text{OP2-2: } \max_{\mathbf{Q}} R_B(\mathbf{Q} | P, \eta_B) \quad (26a)$$

$$\text{s.t. } (22b), (22c)$$

$$0 \leq x_q \leq y_r \quad (26b)$$

$$0 \leq y_q \leq y_r \quad (26c)$$

To enhance reception of Bob while restricting detection of Willie, the optimal position of the reflected spot is Bob's position. Therefore, the solution of OP2-2 can be expressed as $\mathbf{Q}^* = [x_B, y_B, z_B]$ and the corresponding optimal solution γ^* and ω^* of OP2-2 can be calculated according to (24).

After obtaining optimal γ^* and ω^* , we can further decouple OP2 into OP2-3 and OP2-4. Since the covertness constraint is determined by the observations on Willie's side, constraint (22b) is independent to the parameter of Bob's LC-RIS-aided receiver, i.e., η_B . Hence, the decoupled subproblem OP2-3 on power P can be expressed as

$$\text{OP2-3: } \max_P R_B(P | \eta_B, \gamma^*, \omega^*) \quad (27)$$

$$\text{s.t. } (22b)$$

According to (18), we can relax constraint (22b) in OP2-3 as

$$\begin{aligned} & \frac{(4\pi - e) \left[h_W^{LC,*} \left(h_W^{LoS} + \sum_{k=1}^{N_m} h_{W,k}^{mRIS} \right) \xi P \right]^2}{4\pi \sigma_W^2 \ln 2} \\ & + \frac{\left[h_W^{LC,*} \left(h_W^{LoS} + \sum_{k=1}^{N_m} h_{W,k}^{mRIS} \right) \xi P \right]^4 e^2}{16\pi^2 \sigma_W^4 \ln 2} \leq \frac{2\varepsilon^2}{N \ln 2}, \end{aligned} \quad (28)$$

where $h_W^{LC,*}$ denotes h_W^{LC} with optimal η_W^* , which is obtained by OP1-1. By simplifying (28), we obtain the upper bound on P , which is expressed as

$$P \leq \frac{2\pi \sigma_W^2 (4\pi - e) \left[\left(1 + \frac{8\varepsilon^2 e^2}{N(4\pi - e)^2} \right)^{\frac{1}{2}} - 1 \right]^{\frac{1}{2}}}{\xi^2 e^2 (h_W^{LC,*})^2 \left[h_W^{LoS} + \sum_{k=1}^{N_m} h_{W,k}^{mRIS} \right]^2} \triangleq P^*. \quad (29)$$

Inequality (29) indicates the maximum average optical intensity can be used by Alice to satisfy the covertness constraint (22b). Hence, the upper bound in (29) can be treated as the solution of OP2-3, i.e., P^* .

TABLE I
SIMULATION PARAMETERS

Parameter	Value
FoV of PD, ψ_F	80°
Dimming target, ξ	1
Peak power constraint, A	140 mA
Effective physical area of PD, ζ	1 cm ²
Voltage constant of LC-RIS, v_0	0.8 V
Threshold voltage of LC-RIS, v_{th}	1.34 V
Electro-optical conversion coefficient, Γ	12 nm/V
Lower constraint on refractive index, η_1	1.5
Upper constraint on refractive index, η_2	1.7
Responsivity of the PD, ϖ	0.54 A/W
Refraction index of lenses in PD, a	1.5
Refractive index of air, η_a	1
Reflection efficiency of mirrors, ρ	0.9

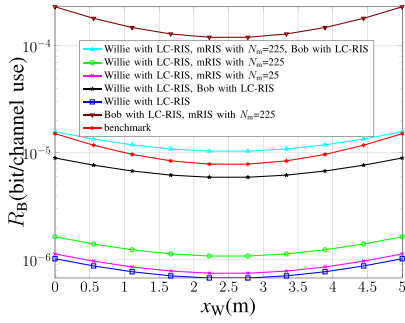


Fig. 2. Achievable rate, R_B , versus x_W with different deployments of RIS when $N = 1000$, and $\varepsilon = 0.1$.

Further, with optimal γ^* , ω^* and P^* , the decoupled subproblem OP2-4 on η_B is expressed as

$$\begin{aligned} \text{OP2-4: } \max_{\eta_B} \quad & R_B(\eta_B | P^*, \gamma^*, \omega^*) \\ \text{s.t.} \quad & (22c) \end{aligned} \quad (30)$$

According to (20), maximizing R_B is equivalent to maximizing the channel gain of Bob. Hence, OP2-5 is rewritten as

$$\begin{aligned} \text{OP2-5: } \max_{\eta_B} \quad & P^* h_B^{LC} \left[h_B^{LoS} + \sum_{k=1}^{N_m} h_{B,k}^{mRIS}(\gamma^*, \omega^*) \right] \\ \text{s.t.} \quad & (22c) \end{aligned} \quad (31)$$

Similar to OP1-1, OP2-5 is a convex problem, which can be solved quickly by the gradient descent method.

V. NUMERICAL RESULTS

In this section, the simulation results are provided to show the feasibility of the optical RIS-aided covert VLC system. The size of the room is $5 \times 5 \times 5$ m³. Alice is in the center of the ceiling. The main simulation parameters are summarized in Table I. The numerical results show that the gradient descent method can be converged within 10-30 iterations.

In the covert VLC system, achieving covertness needs a significant reduction in transmitted power, which leads to a small achievable covert transmission rate R_B [18], [19], [20]. Fig. 2 shows the enhancement of R_B with different deployments of optical RIS. As a benchmark, we consider the case of the covert VLC system without any optical

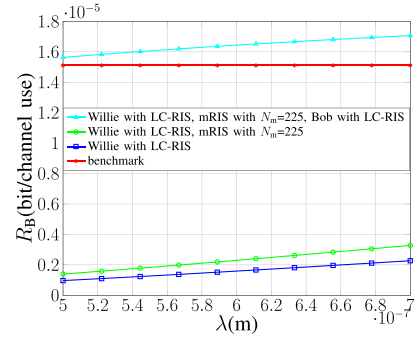


Fig. 3. Achievable rate, R_B , versus wavelength, λ , with different deployments of RIS when $N = 1000$, and $\varepsilon = 0.1$.

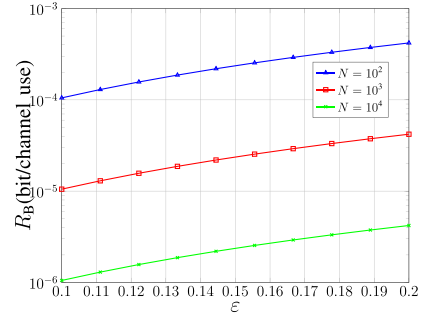


Fig. 4. Achievable rate, R_B , versus hidden constraint value, ε , with different N when $\lambda = 510$ nm.

RIS. When Willie employs an optimized LC-RIS-assisted receiver, the detection ability is improved, which results in the worst case for the covert VLC system. To defense an LC-RIS-assisted Willie, the transmitted power of Alice will be reduced, which results in the decrease of the covert transmission rate of the overall system and R_B is lower than the benchmark. To show the covertness enhancement brought by mRIS, we consider two cases including mRIS with $N_m = 25$ and $N_m = 225$. Fig. 2 shows that an mRIS with insufficient mirrors can not provide a positive effect to combat an LC-RIS-aided Willie. To further enhance the covertness, we can deploy an optimized LC-RIS on Bob's receiver. As shown in Fig. 2, for the worst case, the deployment of LC-RIS on Bob's receiver has a bigger impact on covert VLC than the deployment of mRIS. Moreover, when we jointly optimize the parameters of mRIS and Bob's LC-RIS, the covertness of the VLC system can be further enhanced.

Fig. 3 illustrates the effect of wavelength λ on R_B . Consisting of spectral mirrors, the channel gain brought by the mRIS does not change with λ . For the LC-RIS, according to (9) and (13), the channel gain decreases with λ increasing. Hence, when Willie employs an optimized LC-RIS-assisted receiver, a large λ causes a small h_W^{LC} , which leads to a weak detection and high R_B . When Bob also employs an optimized LC-RIS-assisted receiver, an increase in λ results in a decrease in h_W^{LC} and h_B^{LC} . Since we optimize on the system level, the average optical intensity P increases when h_W^{LC} decreases to achieve higher R_B . Therefore, the positive effect brought by the increase of P and the decrease of h_W^{LC} is dominant, which is greater than the negative effect brought by the decrease of h_B^{LC} . Hence, R_B increases with λ when both Bob and Willie employ an LC-RIS-aided receiver.

Fig. 4 illustrates the change of R_B with ε , which indicates the degree of strictness of the covertness constraint. Particularly, a small ε indicates

a strict covertness constraint. Since the transmission rate needs to be sacrificed to guarantee a strict covertness constraint, R_B decreases with ε decreasing.

VI. CONCLUSION

In this paper, we investigate the optical RIS-aided covert VLC system. Considering both mRIS and LC-RIS, we propose an optimization framework to achieve the maximum covert transmission rate in the worst case. Simulation results show that the deployment of LC-RIS on Bob's receiver has a more significant impact on enhancing covertness compared to the deployment of mRIS. Future work will focus on optimizing scenarios involving multiple PDs, colluding wardens, MIMO, and MISO systems, and exploring RIS strategies to minimize detection probability in these more complex settings.

REFERENCES

- [1] J.-Y. Wang, L.-H. Hong, N. Liu, H.-N. Yang, P. Feng, and J. Ren, "Secrecy analysis and optimization for IRMA-and jammer-aided visible light communications," *IEEE Wireless Commun. Lett.*, vol. 13, no. 7, pp. 1908–1912, Jul. 2024.
- [2] V. Shahiri, H. Behroozi, A. Kuhestani, and K.-K. Wong, "Reconfigurable-intelligent-surface-assisted secret key generation under spatially correlated channels in quasi-static environments," *IEEE Internet Things J.*, vol. 11, no. 9, pp. 15808–15822, May 2024.
- [3] S. Cho, G. Chen, and J. P. Coon, "Physical layer security in visible light communication systems with randomly located colluding eavesdroppers," *IEEE Wireless Commun. Lett.*, vol. 7, no. 5, pp. 768–771, Oct. 2018.
- [4] S. Ma et al., "Optimal probabilistic constellation shaping for covert communications," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 3165–3178, 2022.
- [5] M. Forouzes, F. Samsami Khodadad, P. Azmi, A. Kuhestani, and H. Ahmadi, "Simultaneous secure and covert transmissions against two attacks under practical assumptions," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10160–10171, Jun. 2023.
- [6] J.-Y. Wang, D.-P. Su, P. Feng, N. Liu, and J.-B. Wang, "Optimal height of UAV in covert visible light communications," *IEEE Commun. Lett.*, vol. 27, no. 10, pp. 2682–2686, Oct. 2023.
- [7] Z.-W. Ye, Y.-Y. Zhang, C. Wang, P.-H. Fan, and Y.-J. Zhu, "Ris-aided dynamically adaptive wide field-of-view receiver for visible light communication in industrial Internet of Things," *Opt. Exp.*, vol. 31, no. 21, pp. 34748–34763, Oct. 2023.
- [8] Z.-w. Ye, D. Li, C. Wang, and Y.-j. Zhu, "Ris-assisted optical receiver model and reflected angle optimization for visible light communication in mobile environments," *Opt. Exp.*, vol. 32, no. 13, pp. 22970–22985, Jun. 2024.
- [9] Q. Wu, J. Zhang, Y.-Y. Zhang, and G. Xin, "Asymptotic capacity maximization for MISO visible light communications with mirror array-based RISs on walls and an LC RIS-based receiver," *Opt. Commun.*, vol. 561, Jun. 2024, Art. no. 130496.
- [10] A. R. Ndjiongue, O. A. Dobre, and H. Shin, "Maximal transmission rate in omni-DRIS-assisted indoor visible light communication systems," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13956–13961, Sep. 2024.
- [11] A. M. Abdelhady, A. K. S. Salem, O. Amin, B. Shihada, and M.-S. Alouini, "Visible light communications via intelligent reflecting surfaces: Metasurfaces vs mirror arrays," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 1–20, 2021.
- [12] A. M. Abdelhady, O. Amin, A. K. S. Salem, M.-S. Alouini, and B. Shihada, "Channel characterization of IRS-based visible light communication systems," *IEEE Trans. Commun.*, vol. 70, no. 3, pp. 1913–1926, Mar. 2022.
- [13] L. Qian, X. Chi, L. Zhao, and A. Chaaban, "Secure visible light communications via intelligent reflecting surfaces," in *Proc. IEEE Int. Conf. Commun.*, 2021, pp. 1–6.
- [14] S. Sun, F. Yang, J. Song, and Z. Han, "Optimization on multiuser physical layer security of intelligent reflecting surface-aided VLC," *IEEE Wireless Commun. Lett.*, vol. 11, no. 7, pp. 1344–1348, Jul. 2022.
- [15] S. Aboagye, A. R. Ndjiongue, T. M. N. Ngatched, and O. A. Dobre, "Design and optimization of liquid crystal RIS-based visible light communication receivers," *IEEE Photon. J.*, vol. 14, no. 6, Dec. 2022, Art. no. 7355607.
- [16] O. Maraqa and T. M. N. Ngatched, "Optimized design of joint mirror array and liquid crystal-based RIS-aided VLC systems," *IEEE Photon. J.*, vol. 15, no. 4, Aug. 2023, Art. no. 7303211.
- [17] Q. Zhang, F. Yang, Z. Liu, S. Sun, H. Zhang, and J. Song, "Hardware implementation of a mirror array-based optical intelligent reflecting surface for VLC: Prototype and experimental results," *Opt. Exp.*, vol. 32, no. 11, pp. 19252–19264, May 2024.
- [18] D.-P. Su, D.-S. Lu, Y.-C. Yu, and J.-Y. Wang, "Joint optimization of UAV's height and peak optical intensity in weather-dependent covert VLC," *IEEE Signal Process. Lett.*, vol. 31, pp. 2315–2319, 2024.
- [19] J.-W. Shi, X.-H. Wang, P.-F. Yu, and J.-Y. Wang, "Fundamental performance limit of covert free-space optical communications," *Phys. Commun.*, vol. 54, Oct. 2022, Art. no. 101821.
- [20] Q. Hu, T. Lin, T. Wei, N. Huang, Y.-J. Zhu, and C. Gong, "Covert transmission in water-to-air optical wireless communication systems," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 4432–4447, 2024.